

AI-Based Financial News Sentiment Analysis for Stock Market Prediction

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Abstract— This paper presents an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that combines financial news with historical stock market data to predict stock price direction. The system uses FinBERT for financial sentiment analysis and extracts 35 engineered features covering sentiment dynamics, market momentum, RSI, MACD, volatility, volume behavior, and sentiment-market interactions. These features are integrated using the proposed FinSent-Momentum Fusion (FSMF) framework and used with a Gradient Boosting classifier to predict whether the stock market will move UP or DOWN. The system is evaluated on the TRACE AAPL dataset, where the proposed approach achieves 69.12% accuracy on the held-out test period. An interactive Streamlit dashboard further provides sentiment analysis and stock movement predictions to support data-driven financial decision-making.

Keywords— Financial News, Sentiment Analysis, FinBERT, Stock Market Prediction, NLP, Feature Engineering, Gradient Boosting, FSMF

I. INTRODUCTION

Financial news plays an important role in influencing investor sentiment and stock market movements. Large volumes of financial news are published every day, making manual analysis difficult, time-consuming, and challenging to perform consistently. News articles can contain positive, negative, or neutral information that may influence market trends. Natural Language Processing (NLP) and machine learning techniques can analyze this information and identify meaningful sentiment patterns.

This project proposes an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that combines financial news analysis with historical stock market data. The system uses FinBERT to extract positive, negative, and neutral sentiment from financial news and applies feature engineering to generate 35 features involving sentiment dynamics, market momentum, RSI, MACD, volatility, volume behavior, and sentiment-market interactions.

The proposed FinSent-Momentum Fusion (FSMF) framework combines these features and uses a Gradient Boosting classifier to predict stock movement as UP or DOWN. An interactive Streamlit dashboard is also developed to display sentiment analysis and prediction results.

The system aims to support data-driven financial analysis by combining textual news information with historical market behavior.

II. EASE OF USE

The proposed AI-Based Financial News Sentiment Analysis for Stock Market Prediction system is designed to provide a structured and accessible workflow for analyzing financial news and predicting stock movement. The system separates data collection, preprocessing, sentiment analysis, feature engineering, classification, and visualization into clearly defined stages.

Users do not need to manually calculate sentiment scores or technical indicators. Financial news is automatically processed using FinBERT, while the feature engineering pipeline

prepares the information required by the Gradient Boosting classifier.

The Streamlit application provides an interactive interface through which users can enter financial news and observe the resulting sentiment information and predicted stock movement.

A. Selecting a Template

The research paper is organized using an A4 two-column academic layout. The document contains a centered title, author information, abstract, keywords, numbered sections, subsections, tables, equations, acknowledgment, and references.

The document uses the standard LaTeX article class while reproducing the visual characteristics of the provided two-column research-paper template. Consistent margins, column spacing, typography, and heading styles are maintained throughout the document.

B. Maintaining the Integrity of the Specifications

A consistent formatting structure is maintained throughout the paper. Major sections are represented using uppercase Roman numerals, while subsections use uppercase alphabetic labels.

The body text is fully justified so that the text aligns consistently with both the left and right margins of each column. Paragraph spacing, indentation, font size, and line spacing are kept consistent throughout the document.

Figures and tables are placed close to the relevant discussion, and equations are numbered consecutively.

III. PREPARE YOUR PAPER BEFORE STYLING

The project content was organized into financial news collection, historical market data collection, preprocessing, sentiment analysis, feature engineering, model training, evaluation, and application development.

Financial news is aligned with historical AAPL market information using corresponding dates. FinBERT is then applied to obtain sentiment scores, while market indicators are calculated from historical stock data.

The resulting information is represented using 35 engineered features and provided to the proposed FSMF framework and Gradient Boosting classifier.

A. Abbreviations and Acronyms

The following abbreviations are used throughout the paper:

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **NLP** – Natural Language Processing
- **FinBERT** – Financial Bidirectional Encoder Representations from Transformers
- **FSMF** – FinSent-Momentum Fusion
- **RSI** – Relative Strength Index
- **MACD** – Moving Average Convergence Divergence
- **AAPL** – Apple Inc. stock ticker

Abbreviations are defined when they are first introduced in the paper.

B. Units

The project uses numerical market and sentiment information. Stock prices are represented using their corresponding market currency, while trading volume represents the number of shares traded.

Sentiment outputs are represented as probability or score values associated with positive, negative, and neutral classes. The prediction target is represented as a binary value, where 1 represents UP movement and 0 represents DOWN movement.

Dates are used to align financial news with corresponding historical market observations.

C. Equations

The stock return is represented as:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (1)$$

where P_t represents the stock price at time t and P_{t-1} represents the previous stock price.

The prediction target is defined as:

$$y_t = \begin{cases} 1, & P_{t+1} > P_t \\ 0, & P_{t+1} \leq P_t \end{cases} \quad (2)$$

where $y_t = 1$ represents an UP movement and $y_t = 0$ represents a DOWN movement.

FinBERT generates three sentiment scores:

$$S_t = [S_{positive}, S_{negative}, S_{neutral}] \quad (3)$$

The combined feature representation is:

$$F_t = [S_t, R_t, V_t, I_t, Vol_t, Int_t] \quad (4)$$

where S_t represents sentiment features, R_t represents market momentum, V_t represents technical indicators, I_t represents volatility, Vol_t represents volume behavior, and Int_t represents sentiment-market interactions.

Classification accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (5)$$

Some Common Mistakes

Several issues should be avoided while developing and evaluating a financial prediction system.

Financial news must be correctly aligned with the corresponding trading dates. Incorrect date alignment may associate news with the wrong market observations.

Information from the future should not be included while constructing features for a past prediction. Such data leakage can produce artificially high evaluation results.

Chronological ordering should be preserved when working with time-dependent financial data.

FinBERT should not be interpreted as the stock prediction model. FinBERT performs financial sentiment analysis, whereas Gradient Boosting performs the final UP/DOWN classification.

The 35 engineered features should not be interpreted as 35 different machine learning algorithms. They represent different types of information extracted from financial news and market data.

IV. USING THE TEMPLATE

The paper is organized using a hierarchical structure consisting of major sections and subsections. Each section contributes to explaining the financial news sentiment analysis and stock movement prediction system.

The major technical components include the proposed system, methodology, experimental setup, results and discussion, Streamlit application, limitations and future scope, and conclusion.

A. Identify the Headings

Headings are used to organize the different components of the research paper. Major sections use uppercase Roman numerals, such as

Introduction, Ease of Use, Prepare Your Paper Before Styling, Using the Template, Proposed System, Methodology, Experimental Setup, Results and Discussion, Streamlit Application, Limitations and Future Scope, and Conclusion.

Subsections use alphabetic labels to organize related topics within a major section.

B. Figures and Tables

Figures and tables are important for presenting the architecture, workflow, feature categories, and experimental results of the proposed system.

The proposed workflow consists of financial news and historical AAPL data collection, preprocessing and date alignment, FinBERT sentiment analysis, feature engineering, FinSent-Momentum Fusion, Gradient Boosting, and UP/DOWN prediction.

V. PROPOSED SYSTEM

The proposed system combines financial news sentiment with historical stock market information to predict stock price direction.

The complete workflow consists of financial news and historical AAPL data collection, preprocessing and date alignment, FinBERT sentiment analysis, feature engineering, FSMF feature fusion, Gradient Boosting classification, and Streamlit visualization. The system therefore combines textual information from financial news with numerical information obtained from historical market data.

VI. METHODOLOGY

A. Data Collection

The system uses financial news from the TRACE ACL18 AAPL dataset along with historical AAPL stock market data. The financial news is associated with corresponding dates so that textual sentiment can be analyzed together with market behavior.

Historical market data provide numerical information such as stock prices and trading volume, while financial news provides textual information for sentiment analysis.

B. Data Preprocessing

The collected data undergo preprocessing before model training.

Financial news is cleaned and prepared for sentiment analysis. Historical market data are organized chronologically, and news records are aligned with their corresponding trading dates.

The prediction target is generated using subsequent stock movement. Care is taken to avoid data leakage between training and test periods.

C. Financial Sentiment Analysis

FinBERT is used to analyze financial news because it is designed for financial-domain sentiment analysis. For each news item, FinBERT produces sentiment scores corresponding to positive, negative, and neutral classes. These sentiment scores provide a numerical representation of the information contained in financial news.

D. Feature Engineering

A total of 35 engineered features are used in the proposed system.

These features represent multiple dimensions of financial information, including sentiment, sentiment dynamics, market momentum, RSI, MACD, volatility, volume behavior, and sentiment-market interactions.

The purpose of feature engineering is to provide the machine learning model with a richer representation of both financial news and market behavior.

E. FinSent-Momentum Fusion

FinSent-Momentum Fusion (FSMF) is the proposed

framework for combining financial sentiment information with market-related features.

FSMF is not a separate machine learning algorithm. It provides a structured feature representation that combines multiple sources of information before classification.

The feature vector can be represented as:

$$\mathbf{X} = [\mathbf{S}, \mathbf{D}, \mathbf{M}, \mathbf{R}, \mathbf{V}, \mathbf{I}] \quad (6)$$

The combined representation is then provided to the Gradient Boosting classifier.

F. Gradient Boosting Classification

Gradient Boosting is used as the final machine learning classifier.

The classifier receives the engineered feature representation and predicts the subsequent stock movement.

The two output classes are:

- UP (1)
- DOWN (0)

The model is evaluated using the held-out TRACE AAPL test period.

VII. EXPERIMENTAL SETUP

The proposed system is evaluated using the TRACE AAPL dataset together with historical AAPL market data. A baseline Logistic Regression model is used for comparison. The baseline uses basic sentiment and market features. The proposed system uses 35 engineered features integrated through the FSMF framework and classified using Gradient Boosting.

Table 1: Model Performance

Model	Features	Accuracy
Logistic Regression	Basic sentiment and market features	56%
FSMF + Gradient Boosting	35 engineered features	69.12%

VIII. RESULTS AND DISCUSSION

The baseline Logistic Regression model achieved an accuracy of 56% on the test data.

The proposed FSMF framework combined with Gradient Boosting achieved an accuracy of 69.12% on the held-out TRACE AAPL test period.

The difference between the two reported accuracies is 13.12 percentage points.

Table 2: Feature Categories

Category	Purpose
Sentiment	News polarity
Sentiment Dynamics	Sentiment changes
Momentum	Market movement
Technical Indicators	RSI and MACD
Volatility	Market behavior
Volume Interactions	Trading activity Sentiment-market relationships

The results show that the proposed feature representation provides the Gradient Boosting classifier with multiple sources of information rather than relying only on basic sentiment information.

The reported 69.12% accuracy applies to the held-out TRACE AAPL test period used in this study.

IX. STREAMLIT APPLICATION

An interactive Streamlit application was developed to provide a simple interface for the proposed system.

The user enters financial news into the application. The input text is processed by FinBERT to obtain sentiment information.

The feature processing pipeline then prepares the required features, which are passed to the trained Gradient Boosting classifier.

The application displays the input financial news, positive sentiment score, negative sentiment score, neutral sentiment score, and predicted stock movement.

The final prediction is displayed as either UP or DOWN.

X. LIMITATIONS AND FUTURE SCOPE

The current system is evaluated using AAPL-related data and therefore the reported performance should be interpreted within the scope of the dataset and test period.

The prediction quality depends on the quality, cover-age, and temporal alignment of financial news and historical market data.

The system focuses on binary stock movement prediction rather than directly predicting the exact future stock price.

Future work can extend the system to multiple stocks and larger financial datasets. Additional financial news sources can be incorporated to improve data coverage.

More advanced transformer models and temporal deep learning architectures can also be investigated.

The Streamlit application can be extended with real-time news collection, live market information, portfolio-level analysis, historical prediction visualization, and additional financial indicators.

XI. CONCLUSION

This paper presented an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that combines financial news with historical stock market information.

FinBERT is used to extract financial sentiment, while 35 engineered features capture sentiment dynamics, market momentum, technical indicators, volatility, volume behavior, and sentiment-market interactions.

These features are integrated through the proposed

FinSent-Momentum Fusion framework and classified using Gradient Boosting.

The baseline Logistic Regression model achieved 56% accuracy, while the proposed FSMF and Gradient Boosting approach achieved 69.12% accuracy on the held-out TRACE AAPL test period.

The developed Streamlit application provides an interactive interface for financial news sentiment analysis and stock movement prediction.

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